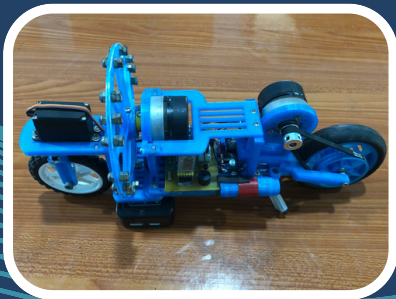
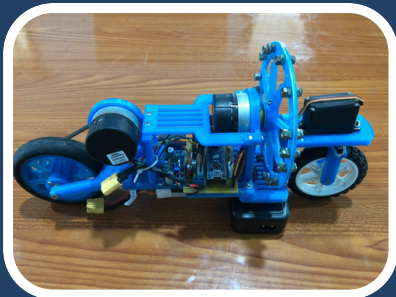


5 Application Areas

- **Robotics and Automation:** Utilized in designing self-balancing mobile robots, autonomous delivery units, and dynamic two-wheeled platforms that require real-time stabilization.
- **Education and Research:** Serves as an experimental platform for teaching and testing control systems engineering, embedded programming, sensor fusion, and PID tuning concepts.
- **Aerospace Stabilization Principles:** Employs the core mechanics of reaction wheels, which share fundamental stabilization and attitude-control principles with satellite and spacecraft orientation systems.
- **Advanced Personal Mobility:** Contributes to the prototyping and research of gyroscopically stabilized bicycles, motorcycles, and narrow electric vehicles designed to prevent tipping and improve safety.



6

Conclusion

In conclusion, this project successfully demonstrates the integration of embedded system and automated control theory to achieve real-time self-balancing using a reaction wheel mechanism. By effectively combining the Arduino Nano, MPU6050 sensor, and PID algorithm, the prototype maintains dynamic stability independently, while the integrated Bluetooth module allows for reliable remote navigation. Ultimately, this system bridges theoretical control concepts with practical mechatronic application, laying a strong foundation for future advancements in automated robotics and personal mobility stabilization.

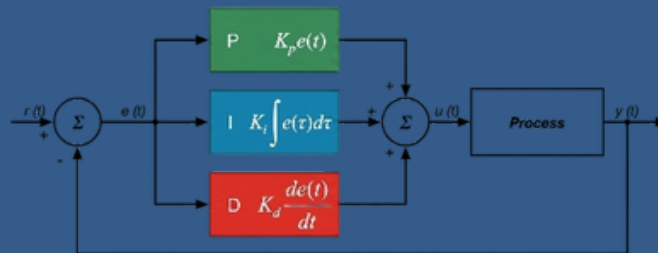


Figure 1.3. PID control system

1



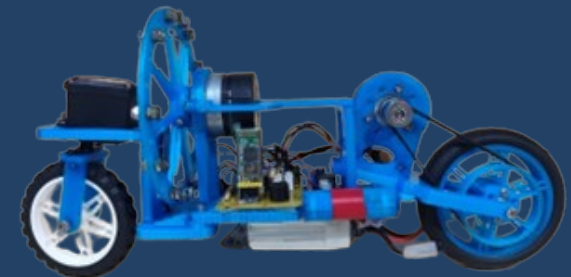
MINISTRY OF SCIENCE AND TECHNOLOGY

TECHNOLOGICAL UNIVERSITY (MANDALAY)

DEPARTMENT OF MECHATRONIC ENGINEERING

PROJECT SHOW & JOB FAIR 2026

PID-BASED REACTION WHEEL DRIVEN
BALANCING BIKE



MG ZAW MYO KHANT

III MC - 1

MG WAI PHYO

III MC - 13

2

Aim

The main aim of this project is to design and implement a self-balancing bike prototype utilizing a PID control algorithm and a reaction wheel mechanism to achieve dynamic balance independently without external supports

Objectives

- ◆ To develop and tune a PID control system integrated with sensor feedback to detect tilt angles and apply real-time corrective torque via the reaction wheel.
- ◆ To assemble and test hardware prototype incorporating the microcontroller, motor, sensors, and reaction wheel to validate its stability and overall performance.

Block Diagram

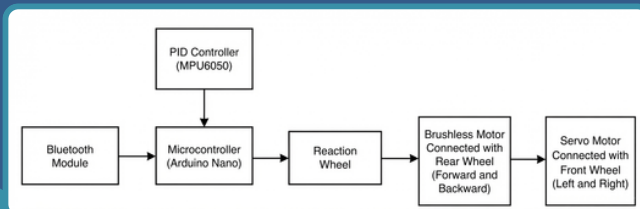


Figure 1.1. Overall Block Diagram of the Bike

3

Operation

The MPU6050 sensor measures the tilt angle of the balancing bike in real-time and sends the data to the Arduino Nano microcontroller. Based on this feedback, the PID control system calculates the required correction and drives the Reaction Wheel to adjust and maintain dynamic balance without external supports. Simultaneously, a user can send wireless commands via the Bluetooth Module to control forward/backward movement and left/right steering.

Flowchart

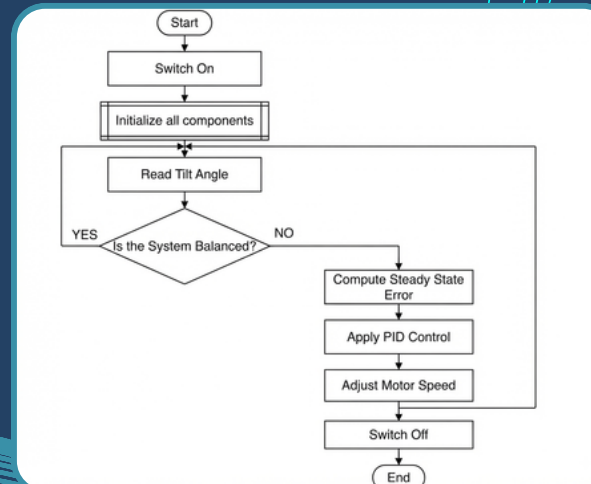
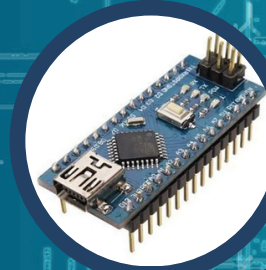


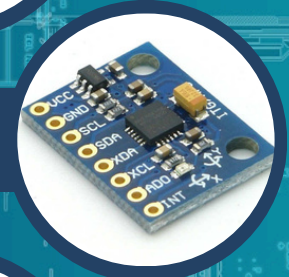
Figure 1.2. Overall System Flowchart of the Bike

4

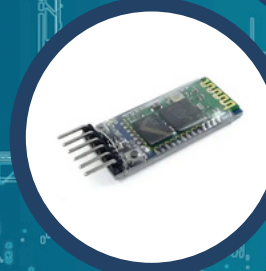
Core Components



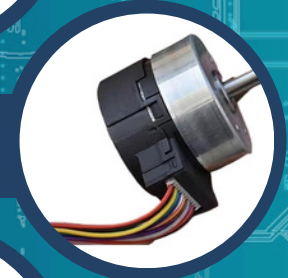
Arduino Nano
Microcontroller



MPU6050
Gyroscope Sensor



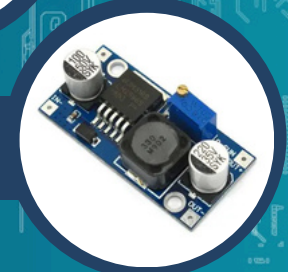
HC-05
Bluetooth Module



Nidec 24H
Brushless Motor



MG-996R
Servo Motor



LM2596
Step-down Converter